Palak Sharma

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Education

M.Sc. Cognitive Science 2019–2021
Centre for Cognitive and Brain Sciences
Indian Institute of Technology Gandhinagar, Gujarat

B.A. Psychology 2015–2018
Fergusson College, Pune, Maharashtra

Research Experience

Exploring the neuro-protective benefits of sensorimotor complexity associated with embodied occupational skills and activities (sub-project under NIH funded project titled "The impact of bilingualism on cognitive reserve/resilience using socio-demographically and linguistically diverse populations")
NIH funded Research Associate, NIMHANS 2025–Present
Supervisor: Dr. Suvarna Alladi

Investigating neuro-protective benefits associated with complex skilled occupations through acquisition, pre-processing and analysis of structural and functional brain scans of demographically matched Indian artisans and controls.

Skill-based Assessments of Cognitive Resilience and Brain Health 2025–2026
ICMR funded Project Research Scientist, NIMHANS
Supervisor: Dr. Suvarna Alladi

Developed a literacy-neutral visual pattern perception task based on simulated weaving drafts to study perceptual decision-making in handloom weaving communities of South India. The project integrates ecological stimulus design, psychophysics, cognitive modeling, and ethnographic fieldwork to examine ecologically grounded perceptual expertise.

Effects of Spatial Reference Frames on Verbal Working Memory 2023–2024
Junior Researcher, Donders Institute, Radboud University
Supervisor: Dr. Pim Haselager

Investigated the influence of visuospatial bootstrapping on working memory by designing a non-word recall paradigm with different spatial reference frame manipulations as conditions, along with eye-tracking integration.

The Effect of Changing Hand Postures on Tactile Subitising 2020–2021
M.Sc. Cognitive Science, IIT Gandhinagar
Supervisor: Dr. Leslee Lazar, Tactile Perception Lab

Investigated embodied foundations of enumeration strategies (subitization, counting, estimation) in tactile perception. Designed a psychophysics experiment using vibrotactile motors and analysed accuracy and reaction time data across parallel-hand, crossed-hand, and interleaved-finger conditions.

Neurophysiological Correlates of Multi-Sensory Perception 2022
Internship, IIT Jodhpur
Supervisor: Dr. Dipanjan Roy, Brain Connectivity, Dynamics and Cognition Lab

Investigated neurophysiological correlates of the McGurk effect using EEG data. Explored audiovisual integration mechanisms underlying speech perception.

Humour Styles and Their Relationship to Resilience 2018
B.A. Psychology, Fergusson College
Supervisor: Himani Raichur

Administered psychological tests to investigate correlations between humour styles and resilience in college students. Examined relationships between self-enhancing and self-deprecating humour styles and resilience measures.

Skills

Research Experimental Design, Psychometrics, Psychological Testing, Data analysis, Scientific Writing.

Programming Python, RStudio.

Methods Eye-tracking, Electrophysiology Analysis, Computational Modelling, Ethnographic Fieldwork, Behavioural experiments, Neuroimaging.

Clinical Neuropsychological test administration - Addenbrooke's Cognitive Examination–III (ACE-III), Clinical Dementia Rating (CDR), Neuropsychiatric Inventory (NPI)

Teaching & Field Engagement Undergraduate Instruction, Academic Mentoring, Community-based Research Engagement and Outreach.

Relevant Coursework

Core Training in Cognitive Science

Computation and Cognition, Perception and Attention, Learning and Memory, Cognitive Neuropsychology, Neuroplasticity, Research Methods in Cognitive Science, Experimental Techniques in Cognitive Science, Philosophy of Mind, Phenomenology, Embodiment and Consciousness, Cognition and Psychoanalysis, Evolutionary Neuropsychology, Creativity and Design, Behavioural Economics.

Specialised Short Courses

Statistical Analysis using Bayesian Inference (IIT Gandhinagar), Seminar on Human Evolution: Of Mosaic, Hybrids and Braids (IIT Gandhinagar), Neurobiology of Everyday Life (Coursera).

Foundational Psychology Training

Abnormal Psychology, Positive Psychology, Organisational Psychology, Research Methods.

Academic Activities

Clinical Neuropsychology

- Conducted neuropsychological assessments for dementia screening at the Neurology Department, NIMHANS (2025).

Teaching & Academic Instruction

- Delivered a presentation on ecologically valid experimental design at an interdisciplinary workshop on weaving research organised by the Handloom Futures Trust, Hyderabad (2025).
- Taught an introductory class on Brain and Cognition for undergraduate Psychology students at Radboud University (2023).

Participation in Academic Events

- Attended the Neuromatch Academy Summer School and completed a project on decoding elbow flexion angles from electrophysiological recordings. (2024)
- Completed NASA's Open Science 101 course, focusing on open and reproducible scientific practices. (2024)
- Attended the Winter School on Cognitive Modelling at the Indian Institute of Technology (IIT) Mandi. (2024)
- Moderated a Lightning Talks session on Language at the Donders Discussions Conference, Radboud University. (2023)
- Presented "Evolving Notions of Meaning in Artificial Intelligence" at the Embodied Intelligence Conference. (2022)
- Attended the Neuromatch Academy Summer School and completed a project on decoding and comparing Neuropixels recordings to classify mouse behaviour. (2022)

Educational Outreach & Community Engagement

- Participated in educational outreach initiatives aimed at engaging rural communities on brain anatomy, psychological and neuroscientific research methods, and the importance of community collaboration in promoting neurological and psychological health and well-being (2025-2026).
- Worked as a communications intern for India Science Fest. (2024)
- Designed game-based instructional methods during an observational internship at New English School, Pune (2018).
- Volunteered at Doorstep School, supporting the educational transition for disadvantaged students (2018).

Video documentation of research

Professionally filmed video documenting fieldwork, qualitative interviews and computer task developed within the ICMR-NIMHANS project. Link - (https://www.youtube.com/watch?v=gnNKMJNvIhU)